Reading the Reading Room

Twelve papers on pneumonia detection from chest X-rays. Accuracy on the **Kermany pediatric CXR benchmark** climbed from 92.8% to 97.6% in six years — and we found that **nobody checked whether the models were looking at the lungs.**

TEAM · 3 PAPERS EACH

Samarth Shekhar
230042
Foundations — slide 3

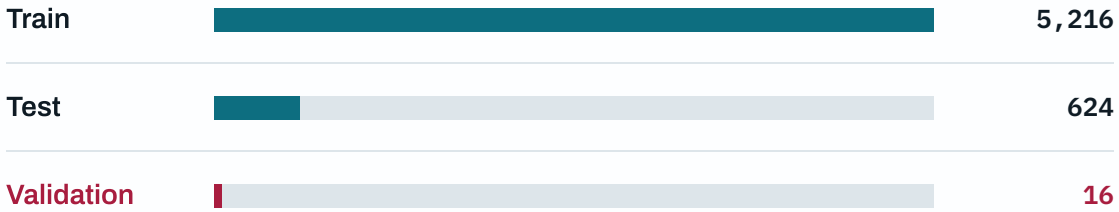
Meghavi Rao
230044
The accuracy race — slide 4

Rachit Kumar
230128
Ensembles & transformers — slide 5

Harshal Nerpagar
230076
The reality check — slide 6

5,856 X-rays. One hospital. Ages 1–5.

Kermany et al. released this set with the 2018 *Cell* paper. It became the default pneumonia benchmark — and its shape explains most of what we found.



The official validation split is **16 images** — 8 per class. It cannot rank two models. In practice every paper tunes against the 624-image test set, then reports on it.

Where it comes from

Guangzhou Women and Children's Medical Center, China. **Pediatric patients aged 1–5.** One site, one scanner fleet, one acquisition protocol, one population.

What the label means

4,273 pneumonia / 1,583 normal — a 2.7 : 1 imbalance. Always predicting "pneumonia" scores 62.5% on the test set, so accuracy alone is a weak claim.

Why the size matters

624 test images gives a 95% confidence interval of roughly **±1.5 percentage points**. A "1% improvement" is six images.

The three papers the whole field is built on

2017–2018: the datasets appear, transfer learning is shown to work, and "radiologist-level" enters the vocabulary.

P1 · BASE PAPER CANDIDATE

Identifying Medical Diagnoses by Image-Based Deep Learning

Kermany et al. · *Cell* 2018 · ~6,100 cites

92.8% acc · 93.2% sens · 90.1% spec

ImageNet-pretrained **Inception-V3**, convolutional layers frozen and the head retrained. Also separated bacterial vs viral pneumonia at 90.7%.

Gave us the dataset and the baseline. **Reports one number, one test set.**

P2 · METHOD

CheXNet: Radiologist-Level Pneumonia Detection

Rajpurkar et al. · arXiv 2017 · Stanford ML Group

F1 0.435 vs radiologists 0.387

121-layer DenseNet on ChestX-ray14. Introduced the Grad-CAM heatmap as the standard way to "explain" a CXR model. One of the few to report **bootstrap 95% CIs**.

Set the architecture template. **An F1 of 0.44 is far from solved.**

P3 · DATASET & SCALE

ChestX-ray8: Hospital-Scale Database and Benchmarks

Wang et al. · CVPR 2017 · NIH

108,948 images · 32,717 patients

Labels **text-mined from radiology reports by NLP**, not read from images — so labels are noisy. Ships bounding boxes and scores localization, not just classification.

The one early paper that measured **where** the model looks.

Everyone swaps the backbone; everyone reports higher

2019–2020: the same 624 test images, a different pretrained network each time. All three papers below use the Kermany data.

P4 · FROM SCRATCH

An Efficient Deep Learning Approach to Pneumonia Classification

Stephen et al. · *J. Healthcare Eng.* 2019 · ~800 cites

CNN trained from scratch + heavy augmentation

Argues you don't need ImageNet at all — a purpose-built CNN plus rotation, zoom and flip augmentation reaches competitive accuracy on 5,856 images.

Augmentation ≈ pretraining, at this scale. **No CI, no external test.**

P5 · HEAD-TO-HEAD

Diagnosis of Pneumonia from Chest X-Ray Images Using Deep Learning

Ayan & Ünver · IEEE EBBT 2019 · ~550 cites

VGG16 87% acc · Xception 82% acc

The useful result is the **split**: VGG16 wins on specificity (91% vs 76%), Xception wins on sensitivity (85% vs 82%). Different networks make different mistakes.

The honest baseline — and the argument for ensembles. **Nobody explains why.**

P6 · BACKBONE SWEEP

Transfer Learning with Deep CNN for Pneumonia Detection

Rahman et al. · *Applied Sciences* 2020 · ~815 cites

DenseNet-201 · 98% normal-vs-pneumonia

Compares AlexNet, ResNet-18, DenseNet-201 and SqueezeNet. Drops to **93.3% on bacterial vs viral** — the clinically useful question is much harder.

Best headline number of the group. **Single split, single seed.**

Three ways to push past 96% on the same 624 images

Combine models, visualise the decision, or replace convolution entirely.

P7 · ENSEMBLE

A Novel Transfer Learning Based Approach for Pneumonia Detection

Chouhan et al. · *Applied Sciences* 2020 · ~890 cites

96.4% acc · 99.62% recall

Ensembles **five** backbones — AlexNet, DenseNet-121, Inception-V3, ResNet-18, GoogLeNet. Highest recall we found, which is the metric that matters for screening.

Confirms Ayan's diversity argument. **5× the compute for +1 point.**

P8 · INTERPRETABILITY

Visualization and Interpretation of CNN Predictions in Pediatric CXR

Rajaraman et al. · *Applied Sciences* 2018 · ~400 cites

Customized VGG16 · 96.2% acc · 93.6% bact/viral

The one Kermany paper that takes explanation seriously — CAM visualisations of what drives each prediction, plus a customised, trimmed VGG16.

Shows heatmaps. **Never scores them against a radiologist.**

P9 · CURRENT SOTA

Efficient Pneumonia Detection Using Vision Transformers on Chest X-Rays

Singh et al. · *Scientific Reports* 2024

97.61% acc · ~95% sens · ~98% spec

Self-attention over image patches instead of convolution, arguing ViT captures **global context** a CNN's local receptive field misses.

The highest number on this benchmark. **Same 624 images, still no CI.**

Three papers that tested whether any of this holds up

Published in *PLOS Medicine* and *Nature Machine Intelligence* — and none of them uses the Kermany dataset. That disconnect is our gap.

P10 · GENERALISATION

Variable Generalization Performance of a Deep Learning Model

Zech et al. · *PLOS Medicine* 2018

AUC 0.931 internal → 0.815 external

Trained on Mount Sinai + NIH, tested at Indiana. Performance collapses off-site. The CNN identified **which hospital** an X-ray came from with 99.95% accuracy, and separated portable from fixed scanners perfectly.

It learned the hospital, not the disease.

P11 · SHORTCUT LEARNING

AI for Radiographic COVID-19 Detection Selects Shortcuts Over Signal

DeGrave, Janizek & Lee · *Nature Mach. Intell.* 2021 · ~980 cites

High in-distribution AUC → fails externally

Names the mechanism. Models key on **laterality markers, image borders, collimation edges and patient positioning** — everything outside the lungs — because those correlate with the label in a single-source dataset.

Exactly the failure a single-hospital dataset invites.

P12 · EXPLANATIONS

Benchmarking Saliency Methods for Chest X-Ray Interpretation

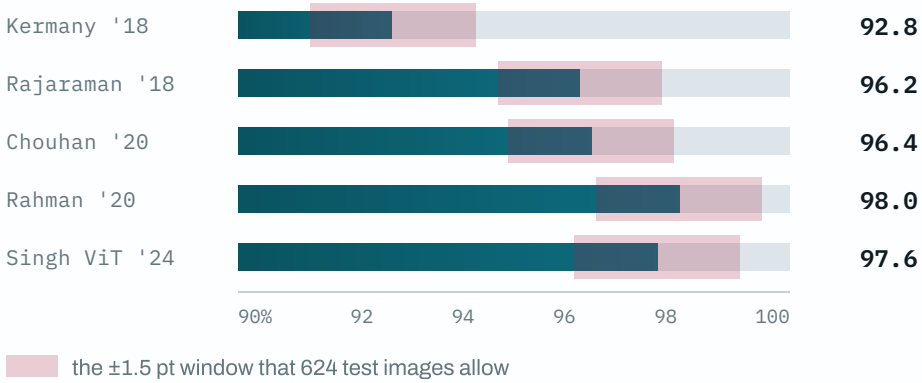
Saporta et al. · *Nature Mach. Intell.* 2022 · CheXlocalize

All 7 saliency methods < human benchmark

Scored seven methods against expert segmentations using mIoU and hit rate. **Grad-CAM was the best of the seven — and still significantly worse than a radiologist.** Worst on small, complex findings.

A convincing heatmap is not evidence.

Accuracy climbed. The evidence behind it didn't.



Headline accuracy as published, all on the same 624-image test set. **Only the 2018 baseline sits clearly outside the rest.** Every gain claimed since — 96.2 through 98.0 — falls inside the shaded window, so it is not separable from sampling noise. No paper reports a confidence interval or a seed variance, so none of them could tell.

PAPER	USES KERMANY	EXTERNAL VALIDATION	CIS OR MULTI-SEED	CALIBRATION REPORTED	LOCALIZATION SCORED
THE NINE BENCHMARK PAPERS					
Kermany '18	●	○	○	○	○
Rajpurkar '17	○	○	●	○	○
Wang '17	○	○	○	○	●
Stephen '19	●	○	○	○	○
Ayan & Ünver '19	●	○	○	○	○
Rahman '20	●	○	○	○	○
Rajaraman '18	●	○	○	○	○
Chouhan '20	●	○	○	○	○
Singh ViT '24	●	○	○	○	○
THE THREE THAT CHECKED					
Zech '18	○	●	●	○	○
DeGrave '21	○	●	●	○	●
Saporta '22	○	○	●	○	●

● reported, benchmark paper ● reported, critique paper ○ not reported

The gap none of the twelve papers closes

> **Every paper on this benchmark reports a single accuracy number on the same 624 single-centre images. Every paper that tested whether such models actually work used a different dataset. Nobody has run the reality check on the benchmark itself.**

GAP 01

Selection on the test set

A `16-image` validation split cannot rank models, so tuning happens against the 624 test images. Reported numbers are optimistic by construction.

GAP 02

No error bars

Zero of the nine benchmark papers report a CI or run multiple seeds. Every gain claimed since 2019 — `96.2 → 98.0` — sits inside the noise window.

GAP 03

No external validation

One hospital, one city, ages 1–5. Zech showed `−0.12 AUC` off-site. Nobody has measured that drop for this dataset.

GAP 04

No calibration

Not one paper reports `ECE` or a reliability curve. A triage tool needs a trustworthy confidence, not just a correct label.

GAP 05

Heatmaps never scored

Grad-CAM appears as figures in most papers. Saporta showed scoring `changes the verdict`. Nobody scores them here.

Reproduce Kermany, then test whether it needs the lungs

M2 · REPRODUCE

Kermany et al. 2018, Inception-V3

Public dataset, ImageNet weights, a clearly stated target: **92.8% acc, 93.2% sens, 90.1% spec.** Small enough to train on a free GPU, well-specified enough to actually match.

Fallback if reproduction stalls: Rahman 2020 (DenseNet-201) — same data, more implementations available.

M3 · OUR ONE CHANGE

Train and test on lung-cropped X-rays

Segment the lung fields and crop. Everything the shortcut literature blames — **laterality markers, collimation edges, borders, positioning** — is outside that crop. Identical model, identical schedule, only the input region changes.

PREDICTION — SUBMITTED BEFORE WE CODE

Standard test accuracy will move **less than 1 point** (no free lunch). But under a context-ablation test — lungs masked out, periphery kept — the full-image model will stay well above the 62.5% majority baseline while the cropped model falls toward it. That gap is the shortcut, measured.

M4 · M5 · MEASURE HONESTLY

The evaluation the field skipped

- **Patient-wise re-split** — a real validation set, not 16 images
- **5 seeds, mean ± 95% CI** on every number we report
- **Sensitivity / recall + AUC**, never accuracy alone
- **ECE + reliability curve** — the first calibration numbers on this benchmark
- **Grad-CAM hit rate inside the lung field**, scored, not shown

Ablations: crop only at test, crop only at train, and full-image control.